

Optical Integrated Sensing and Communication for 6G: A Cross-Modality Survey and Metric-Governed Roadmap

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Abstract—Optical Integrated Sensing and Communication (O-ISAC) is emerging as a 6G design direction because optical carriers can support high-rate links and sensing functions across guided fiber, free-space optical, VLC/LiFi, and photonics-assisted high-frequency platforms. Yet the literature remains fragmented by modality, terminology, measurement plane, and metric semantics, making direct comparison risky without an explicit governance layer.

This systematic review synthesizes 220 peer-reviewed O-ISAC studies using a PRISMA-grounded process and develops a cross-modality taxonomy spanning medium class, integration mechanism, detection/observability regime, and sensing task. We introduce a metric-governed reporting contract that separates bandwidth-limited range resolution $\Delta r_{\min}$, estimator-level accuracy such as σ_r /RMSE, CRB/FIM bound statements, fiber spatial granularity Δz , and optical/electrical signal-quality planes. This contract enables conservative rate–resolution analysis and capacity-resolution quotient (CRQ) synthesis only when rate and sensing metrics are reported under compatible scenario records.

The review further maps enabling technologies, including optical RIS/ORIS, optical phased arrays, photonic integration, photonics-assisted mmWave/THz generation, and ML/security-aware adaptation, and links them to deployment constraints and benchmark needs. The resulting roadmap identifies the reporting practice, taxonomy alignment, reproducible evaluation workflows, scalable hardware, and domain-aware validation needed for credible O-ISAC progress.

Index Terms—Optical integrated sensing and communication, O-ISAC, free-space optics, visible light communication, distributed fiber sensing, photonic-terahertz systems, systematic review.

I. INTRODUCTION

Integrated sensing and communication (ISAC) is now a core 6G design direction because future systems must exchange data and observe physical context in the same workflow [1], [2]. Optical integrated sensing and communication (O-ISAC) extends that premise onto optical carriers and therefore changes both the opportunity and the comparison problem. Fiber links, free-space optical links, VLC/LiFi systems, and photonics-assisted mmWave/THz chains do not share one channel model, receiver plane, or sensing metric vocabulary, even when they all use an O-ISAC label [3]–[5]. Fig. 1 is retained as the opening landscape view because it captures the main physical motivation and modality split.

The practical appeal is clear: photonic-THz demonstrations report wireless rates around 120 Gbps with millimeter-level

range resolution, while multicore fiber experiments report aggregate transmission above 200 Tb/s with concurrent sensing [6], [7]. These results should not be pooled as if they were the same architecture. RF/mmWave ISAC remains constrained by spectrum scarcity, bandwidth-limited sensing, and high-frequency hardware complexity; optical systems open wider spectral resources but split into guided and wireless operating regimes with different impairments, calibration needs, and signal-quality reference planes [3], [8], [9]. The review therefore uses metric-governed comparison rather than headline performance ranking. Frequently used notation and acronyms are provided in the supplementary material.

The fragmentation challenge reduces to four fault lines: terminology aliasing across RadCom/JRC/VLC/photonic/fiber labels; metric non-isomorphism between $\Delta r_{\min}$, RMSE/ σ_r , CRB/FIM, Δz , OSNR, and electrical SNR; modality siloing across fiber, FSO, VLC/LiFi, and photonic-THz communities; and weak cross-domain transfer of waveforms, calibration practice, and benchmarks. This review addresses those faults through five contributions: a PRISMA/TQAF evidence base, a cross-modality taxonomy, a metric-governed reporting contract, a governed rate–resolution and CRQ synthesis, and a compact deployment/enabler/roadmap map. The original RF–O-ISAC comparison table and introductory modality preview figure have been moved to the supplementary file for traceability.

II. BACKGROUND AND METRIC-GOVERNANCE CONTRACT

An O-ISAC system couples an optical source, modulation and waveform resources, a propagation medium, receiver observables, and communication/sensing estimators. The shared abstraction in Fig. 2 is useful only if the modality-specific pieces remain visible: coherent receivers preserve complex-field information, IM/DD receivers expose intensity-domain observations, fiber channels are guided and dispersive/nonlinear, and wireless optical channels add geometry, turbulence, pointing, and ambient-light constraints [7], [14]–[16]. The conceptual NLSE and extended channel derivations are moved to the supplementary file; in the main text, their role is to remind the reader that guided propagation, wireless propagation, and photonic high-frequency generation are not interchangeable channel assumptions.

The protected metric contract is summarized in Table III. Bandwidth-limited range resolution is

$$\Delta r_{\min} = \frac{v}{2B_{\text{eff}}}, \quad (1)$$

with $v = c$ in free space and $v \approx c/n_g$ in guided media when a two-way convention is appropriate. It is not estimator

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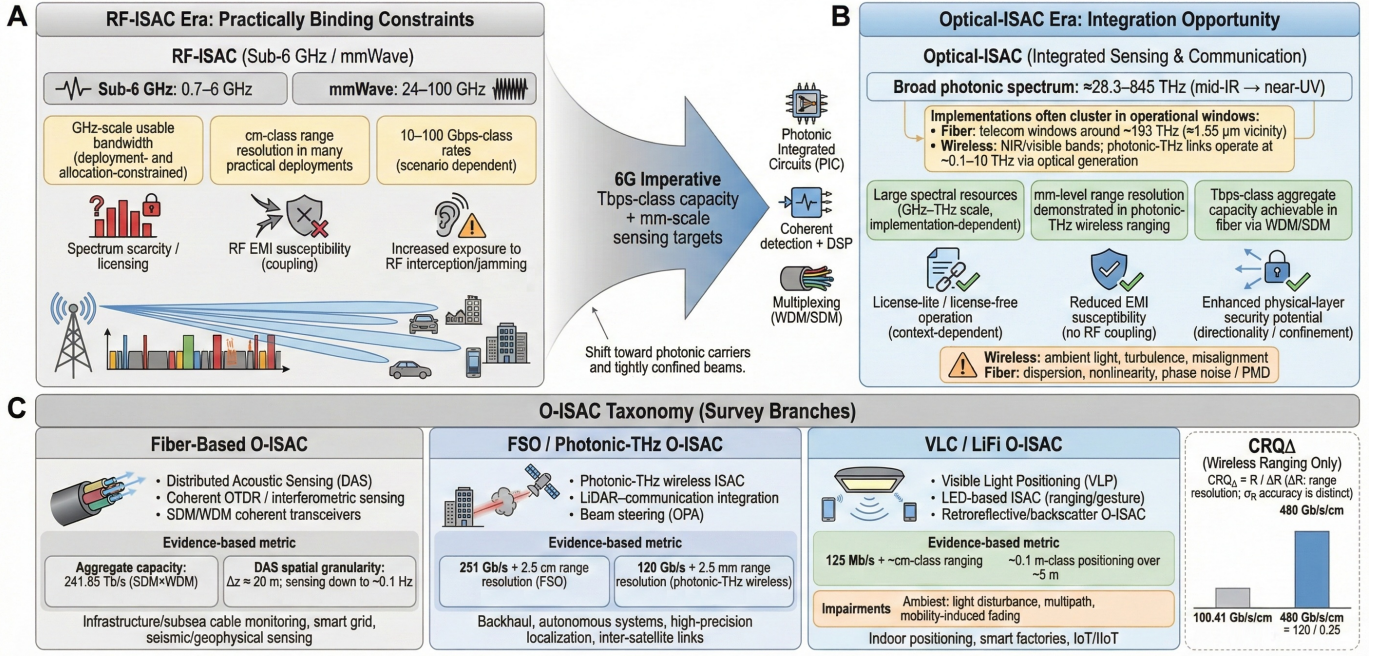


Fig. 1. O-ISAC landscape across optical spectrum resources and the main fiber, FSO/photonic-THz, and VLC/LiFi branches.

TABLE I
RELATED SURVEY LANDSCAPE AND THE CONTRIBUTION OF THIS REVIEW.

Work	Year	Scope	Main emphasis	Syst.	Cross-med.	Tax.	Metric gov.	CRQ
[10]	2025	RF/6G ISAC	RF-centric survey framing and standardization context	No	No	Part	No	No
[4]	2024	VLC	Visible-light JCS prospects and indoor positioning links	No	No	Part	No	No
[11]	2024	VLC/VLP	VLC/VLP channel and localization survey	No	No	Part	No	No
[12]	2024	Fiber	Fiber-sensing ISAC challenges	No	No	Part	No	No
[3]	2024	Optical ISAC	Architecture, potential, and challenge tutorial	No	Part	Part	Part	No
[1]	2026	Photonic THz	Photonic THz-ISAC waveform/system view	No	Part	Part	No	Part
[13]	2025	RIS/ISAC	RIS-enabled ISAC enabler context	No	No	Part	Part	No
This review	2026	All optical	PRISMA $N = 220$, taxonomy, metric contract, governed tradeoff, enablers, applications, roadmap	Yes	Yes	Yes	Yes	Yes

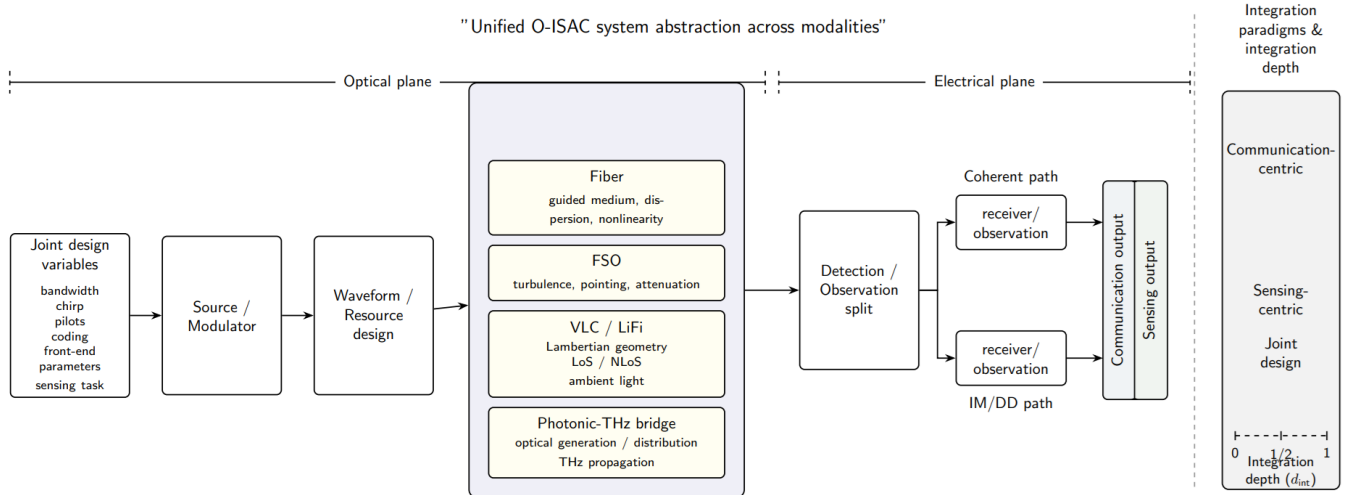


Fig. 2. Unified O-ISAC source-channel-observation abstraction with modality-specific propagation and receiver planes retained.

accuracy. Accuracy terms such as RMSE and σ_r depend on SNR, geometry, estimator choice, and data model. CRB/FIM

statements are bound-level claims under explicit observation assumptions. Fiber spatial granularity Δz is a gauge/segment

TABLE II
MODALITY-AWARE O-ISAC ABSTRACTION.

Modality	Channel basis	Receiver/observable	Typical task	Governance warning
Fiber	Guided dispersive/nonlinear link; DAS/DFOS granularity	Coherent or direct optical/electrical plane	Vibration, strain, intrusion, co-route monitoring	Report Δz and OSNR plane; do not call it wireless range resolution
FSO	LoS optical wireless with turbulence and pointing	Direct or coherent photodetection	Ranging, tracking, link adaptation	Geometry and weather condition admissibility dominate comparison
VLC/LiFi	Lambertian LoS/NLoS indoor optical wireless	Mostly sometimes camera-assisted	IM/DD, gesture, indoor sensing	Electrical SNR/RMSE cannot be merged with optical OSNR or CRB without model disclosure
Photonic-THz/hybrid	Optical generation/distribution plus mmWave/THz or mixed links	Coherent photonic, electronic, or hybrid chains	High-rate ranging and fiber-wireless transfer	Separate optical generation, wireless propagation, and post-detection measurement planes

TABLE III
METRIC CONTRACT FOR ADMISSIBLE O-ISAC COMPARISON.

Metric object	Valid role	Must report	Forbidden shortcut
$\Delta r_{\min}$	Bandwidth-limited resolution	B_{eff} , propagation convention, scenario	Replacing RMSE or CRB with resolution
σ_r /RMSE	Estimator-level accuracy	estimator, SNR plane, geometry, dataset/simulation setup	Treating as bandwidth-only resolution
CRB/FIM	Bound context	observation model, parameters, noise statistics	Treating as measured accuracy
Δz	Fiber spatial granularity	gauge length/segment, interrogator assumptions	Treating as wireless range resolution
OSNR/SNR/ESNR	Signal-quality plane	optical/electrical/reference location and bandwidth	Mixing planes without receiver/noise model
CRQ_{Δ}	Rate per bandwidth-limited resolution	matched R and $\Delta r_{\min}$ in one scenario record	Combining unmatched rate and sensing entries

property rather than a free-space range-resolution substitute. OSNR, post-detection electrical SNR, ESNR, and simulation SNR must remain plane-tagged.

For the governed synthesis, the capacity-resolution quotient is used only as

$$\text{CRQ}_{\Delta} = \frac{R}{\Delta r_{\min}}, \quad (2)$$

and only when the rate and $\Delta r_{\min}$ belong to the same scenario record. The broader optimization view is therefore reduced here to a bridge: Section V interprets performance clouds only after the taxonomy axes and metric roles have been aligned.

III. REVIEW METHODOLOGY

This review follows PRISMA 2020 and PRISMA-S search-reporting principles. The protocol was registered with OSF on February 12, 2026 (Registration ID: 7f6wb), and the formal database search used IEEE Xplore, Scopus, and Web of Science, frozen on November 30, 2025. Exact search strings, search-stage audit material, and moved method-detail tables are preserved in the supplementary evidence package rather than repeated in the main manuscript.

The TQAF appraisal used five compact dimensions: topical fit, metric transparency, modality/implementation clarity, methodological reproducibility, and contribution relevance to cross-modality synthesis. Data extraction recorded medium, integration, detection/observability, sensing task, communication metrics, sensing metrics, signal-quality plane, and scenario compatibility. The final corpus includes 222 full texts

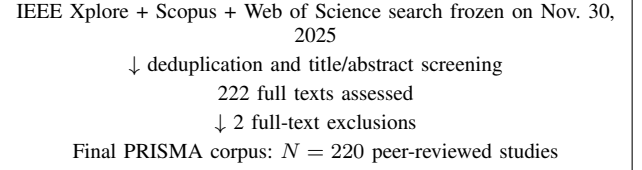


Fig. 3. Compressed PRISMA flow for the final O-ISAC corpus.

assessed, 2 full-text exclusions, and $N = 220$ included studies. The bibliography remains aligned with the included-corpus ledger; a small audit-tail is retained for traceability rather than treated as primary evidence in the narrative.

IV. UNIFIED O-ISAC TAXONOMY

The taxonomy keeps ambiguity visible instead of forcing all papers into a single optical-ISAC mold. Each paper p is mapped as

$$T(p) = (m(p), i(p), d(p), s(p)), \quad (3)$$

where m is medium class, i is integration class, d is detection/observability class, and s is sensing task. The detailed operational contract and the former medium/integration/detection cluster tables are moved to the supplementary file; Table V keeps the main-text synthesis. Counts indicate evidence concentration, not technical superiority.

Fiber O-ISAC is the guided anchor, with 45 records dominated by shared-front-end integration and a task mix spanning

TABLE IV
COMPACT ELIGIBILITY CRITERIA USED FOR STUDY SELECTION.

Criterion	Include	Exclude
Topical scope	Optical, fiber, FSO, VLC/LiFi, photonic-mmWave/THz, or hybrid systems with both sensing and communication relevance	Pure communication, pure sensing, or non-optical RF-only ISAC without optical relevance
Evidence type	Peer-reviewed articles, conference papers, and survey/review works needed for context	Editorial-only, incomplete metadata, inaccessible, or duplicate records outside the audit ledger
Synthesis value	Reports architecture, metric, modality, experiment/simulation, or application information usable for taxonomy/TQAF extraction	No usable O-ISAC extraction fields after full-text assessment

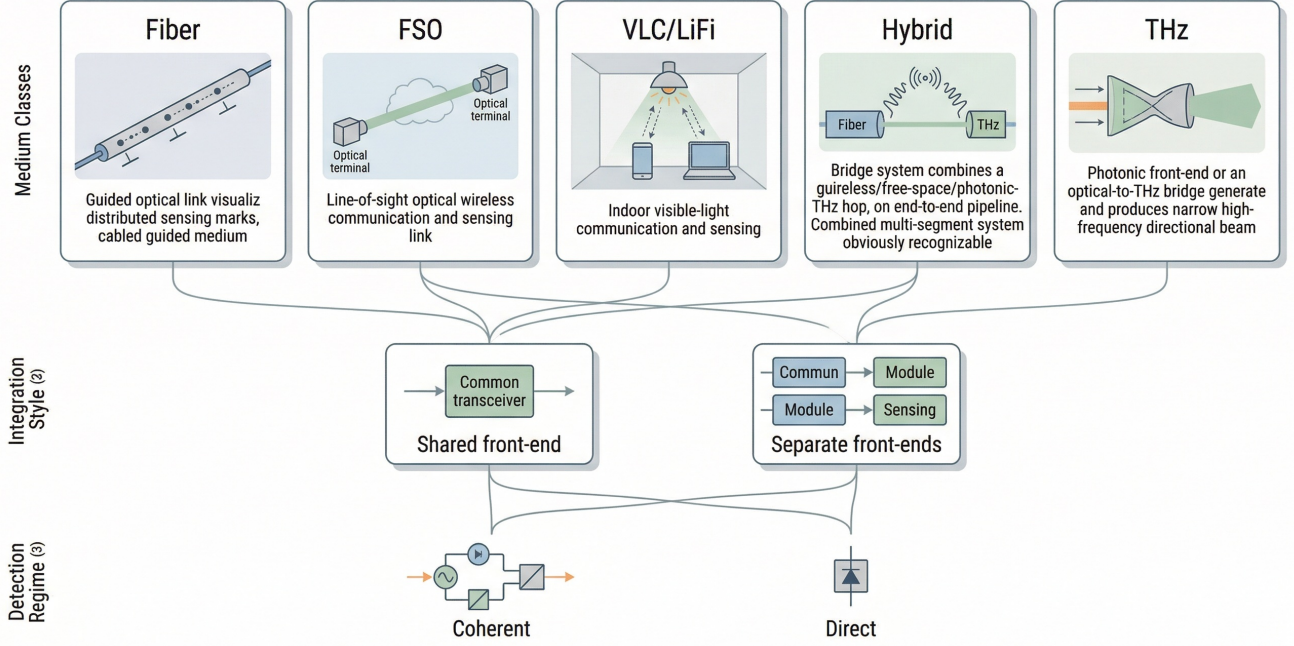


Fig. 4. Semantic taxonomy map retained as the single main-text taxonomy figure.

TABLE V
COMPACT TAXONOMY SUMMARY ACROSS THE $N = 220$ CORPUS.

Medium	Corpus share	Dominant detection	Dominant task	Governance warning
Fiber	45/220	Coherent-leading, with direct tail	Range/vib. monitoring	Compare using guided-channel and Δz /OSNR semantics
FSO	19/220	Mixed direct/coherent	Ranging-heavy	Condition on geometry, turbulence, and pointing assumptions
VLC/LiFi	25/220	Direct IM/DD	Localization and ranging	Electrical-plane accuracy dominates; do not infer optical OSNR behavior
Photonic-THz / hybrid	116/220 hybrid; 39 photonic-THz anchors	Near-balanced coherent/direct	Ranging-heavy with broad multi-task tail	Separate optical generation/distribution from wireless propagation and receiver plane

ranging, vibration, strain, and infrastructure monitoring [7], [12], [17]. FSO is smaller but strongly ranging-oriented, so its evidence should be read through weather, pointing, and geometry constraints [14], [18]. VLC/LiFi is the direct-detection and localization-oriented branch, where Lambertian channels and indoor deployment assumptions govern comparability [4], [15], [19]. Photonic-THz and hybrid systems form the broadest bridge class, linking optical generation/distribution, fiber-wireless transport, and high-frequency wireless sensing/communication [1], [6], [20], [21]. Across all four, tax-

onomy is a conditioning layer for Section V, not a ranking table.

V. COMMUNICATION-SENSING TRADEOFF SYNTHESIS

The governed tradeoff synthesis is the core analytical result and is preserved. Across the $N = 220$ paper corpus, the extraction produced 225 scenario points. Only 20 scenario records support rate plus $\Delta r_{\min}$, only 16 support rate plus σ_r /RMSE, and only 13 support the full rate- $\Delta r_{\min}$ - σ_r triplet. The sparse governed subset is therefore not a data-cleaning

TABLE VI
GOVERNANCE ATTRITION FROM REPORTED METRICS TO ADMISSIBLE SYNTHESIS.

Evidence layer	Available count	Main retained use	Main attrition cause
Scenario records	225	Corpus-level context	Heterogeneous reporting and modality planes
Rate + $\Delta r_{\min}$	20	CRQ-admissible resolution view	Missing matched bandwidth-limited resolution
Rate + σ_r /RMSE	16	Accuracy-conditioned operating view	Missing estimator/SNR/geometry detail
Full triplet	13	Most conservative cross-metric subset	Rate, resolution, and accuracy rarely co-reported

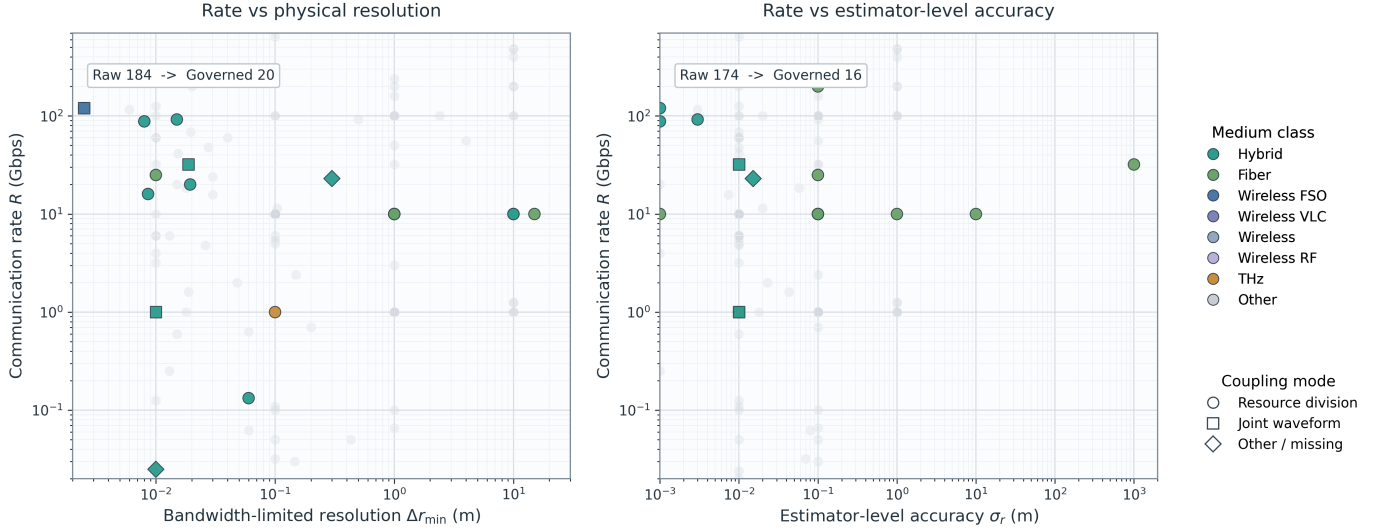


Fig. 5. Governed rate-resolution and rate-accuracy operating clouds after metric-role filtering.

TABLE VII
GOVERNED FIBER, WIRELESS, AND HYBRID SLICES.

Slice	Strong evidence	Weakness for pooling	Main reading
Fiber	Tb/s-class aggregate links and distributed sensing [7]	Δz and OSNR differ from wireless range/ESNR	Guided infrastructure and monitoring anchor
Wireless optical	FSO/VLC ranging/localization and OPA-controlled optical wireless links [14], [22]	Geometry, turbulence, ambient light, and estimator differences	Scenario-conditioned sensing/communication tradeoff
Photonic-THz/hybrid	High-rate photonic generation plus mmWave/THz ranging [6], [20]	Optical and wireless planes can be conflated	Bridge class, not a single medium

accident; it is evidence that O-ISAC reporting remains too heterogeneous for broad rate-sensing frontier claims.

Fig. 5 is therefore the main governed operating-cloud figure. The former sparse CRQ frontier figure is moved to the supplementary file because its correct interpretation is narrow: it illustrates the two nondominated points inside the 20-point CRQ-valid subset, not a stable design envelope for the field. The main conclusion is stronger and safer in text: CRQ is useful only for matched rate and $\Delta r_{\min}$ scenario records, and current reporting practice leaves that subset sparse.

VI. ENABLING TECHNOLOGIES AND SYSTEM-LEVEL CO-DESIGN

Enablers are retained as a compact synthesis because they explain how the taxonomy and metric contract may become deployable systems. Table VIII keeps one main-text enabler table; the broader enabler landscape figure and optimization scaffolding equations are moved to the supplementary file.

For this pass, multi-user resource optimization and secrecy/security optimization equations are no longer numbered in the main text. Their conceptual message remains: enablers become useful only when control variables, impairment models, and benchmark fields are reported together. AI/ML

TABLE VIII
COMPACT ENABLER SYNTHESIS FOR O-ISAC.

Enabler	Main O-ISAC role	Evidence center	Reporting risk
ORIS / optical RIS	Programmable reflection, beam control, spatial reuse	Optical wireless and RIS-adjacent ISAC [13]	Geometry and control overhead must be disclosed
OPA	Beam steering and joint communication/sensing aperture	OPA-based optical wireless ISAC [22], [23]	Beam pattern, scan schedule, and sensing estimator must be linked
PIC / photonic integration	Compact source/modulator/receiver integration	Fiber-wireless and chip-scale demonstrations [24], [25]	Device results need system-level metric planes
Photonics-assisted mmWave/THz	High-frequency generation/distribution and coherent reception	W-band, D-band, and THz-over-fiber systems [1], [20]	Separate optical generation from wireless propagation
ML/security adaptation	Inference, calibration, robustness, and attack-aware control	Learning-assisted VLC/fiber-wireless studies [15], [26]	Dataset, generalization, and threat model must be explicit

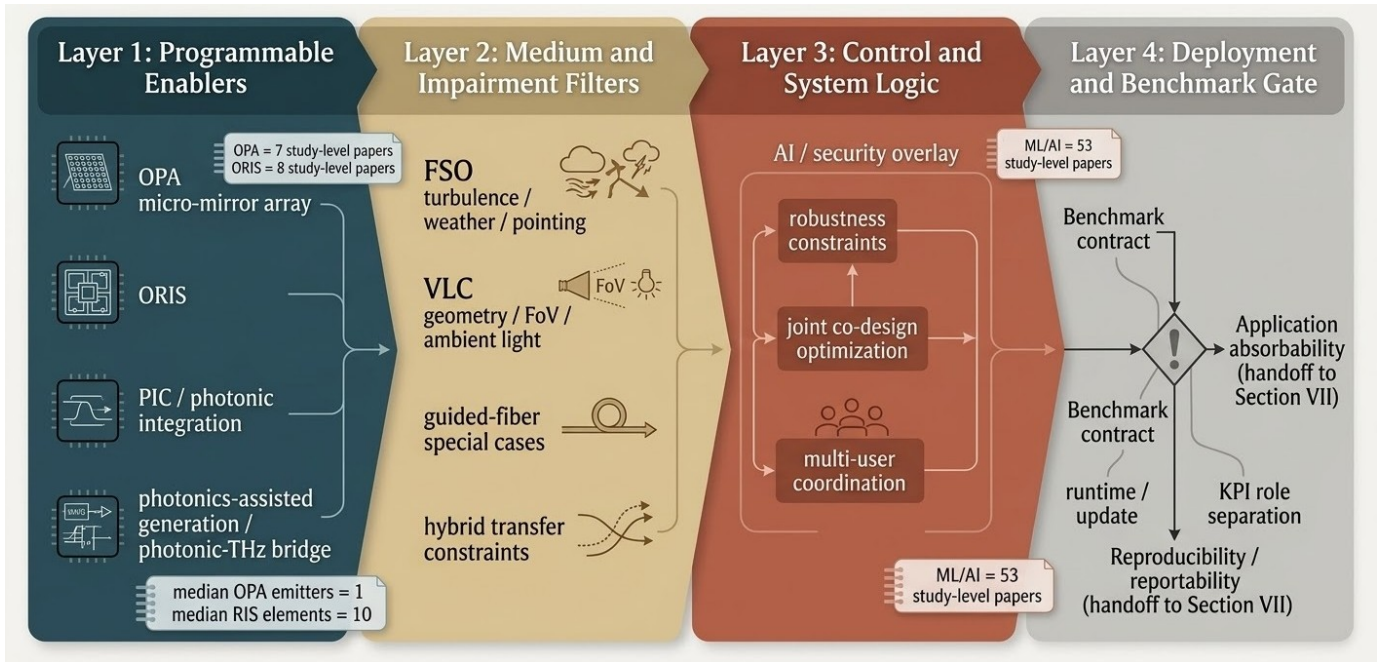


Fig. 6. Deployment-oriented systems map linking optical enablers, impairments, control, and benchmark discipline.

TABLE IX
MINIMUM REPRODUCIBLE REPORTING FIELDS.

Field	Required disclosure
Scenario	medium, geometry, range, mobility, weather/ambient state
Link	rate, BER/throughput method, bandwidth, modulation, multiplexing
Sensing	task, $\Delta r_{\min}$ or Δz , RMSE/ σ_r , CRB/FIM if used
Plane	OSNR/SNR/ESNR reference point and conversion model if any
Evidence	simulation/experiment, data/code availability, uncertainty reporting

is treated as one adaptation layer rather than a standalone maturity claim.

VII. APPLICATIONS AND USE CASES ACROSS DOMAINS

The application evidence is compressed to a deployment map rather than a maturity scorecard. Table X replaces the longer portfolio and audit layer; the original application figure

and dual-view consistency table are preserved in the supplementary file.

Across these domains, the recurring motif is not that one application is mature across all optical media. Instead, each domain inherits a specific observation plane and deployment constraint: fiber corridors inherit Δz and OSNR semantics, indoor VLC/LiFi inherits IM/DD localization assumptions, vehicular optical links inherit geometry and blockage sensitivity, harsh OWC inherits medium-specific attenuation, and satellite/LEO cases inherit pointing and network-integration constraints [27]–[31]. The application map is therefore used to identify where methods may transfer and where the taxonomy should block unsafe analogy.

VIII. OPEN CHALLENGES AND RESEARCH ROADMAP

The roadmap is compressed to one figure, one challenge table, and one five-row agenda. Fig. 7 remains the main

TABLE X
APPLICATION PORTFOLIO COMPRESSED TO FIVE MACRO-DOMAINS.

Macro-domain	Communication motif	Sensing motif	Transfer warning
Smart infrastructure / cabled-fiber corridor	High-capacity transport and monitoring overlays	Vibration, strain, intrusion, seismic/corridor awareness	Strong for guided infrastructure; weak as wireless baseline
Indoor VLC/LiFi localization	Lighting/data access and room-scale links	Positioning, gesture, occupancy, camera-assisted sensing	Direct-detection and ambient-light assumptions dominate
Automotive / optical V2X	Short-range optical links and beam-controlled access	Ranging, pose, cooperative perception	Geometry, eye safety, weather, and blockage must be disclosed
Underwater or harsh OWC	Optical wireless where RF is limited	Salinity, environmental, and link-state monitoring	Medium absorption/scattering makes transfer highly conditional
LEO/satellite or space photonic O-ISAC	Space/NTN optical backhaul and hybrid network support	Tracking, ranging, situational awareness	Evidence is roadmap-oriented and needs deployment validation

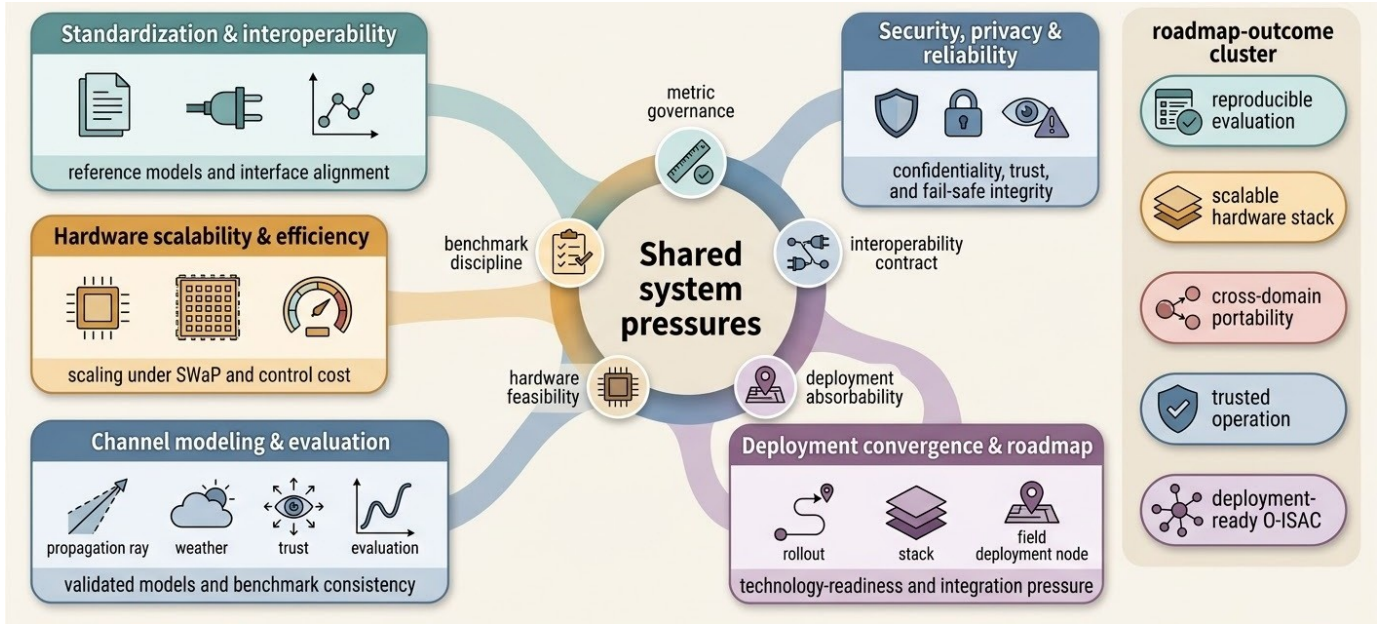


Fig. 7. Challenge-to-roadmap dependency map for taxonomy, metrics, benchmarks, hardware, security, and deployment validation.

roadmap graphic because it links the challenge domains back to metric governance, enablers, and applications.

The removed roadmap utility and governance equations were organizational scaffolding, not necessary technical contributions. The remaining roadmap message is direct: O-ISAC needs interoperable taxonomy, metric-governed reporting, reproducible benchmarks, scalable hardware, and deployment validation.

IX. CONCLUSIONS

This review synthesized optical integrated sensing and communication as a family of architectures rather than a monolithic interchangeable design space. Using a PRISMA-grounded corpus of 220 studies, it organized fiber, FSO, VLC/LiFi, photonic-THz, and hybrid work through a cross-modality taxonomy and a metric-governed comparison contract. The central lesson is that comparability requires explicit modality, metric role, measurement plane, and deployment context: $\Delta r_{\min}$ is not σ_r /RMSE, CRB/FIM is a bound-level statement, fiber Δz is not wireless range resolution,

and OSNR and electrical SNR cannot be exchanged without a receiver/noise model. Under that governed reading, the rate-resolution evidence is informative but sparse: only small subsets of the 225 extracted scenario records support admissible CRQ or full triplet synthesis. Future progress therefore depends as much on reporting practice as on hardware advances. Reproducible workflows, benchmark definitions, scalable OPA/ORIS/PIC and photonic-THz hardware, security-aware adaptation, and domain-aware validation are the necessary path from promising O-ISAC demonstrations to credible 6G systems.

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TABLE XI
COMPRESSED O-ISAC CHALLENGE DOMAINS.

Domain	Bottleneck	Roadmap response
Std./interoperability	Terminology, taxonomy, and reporting fields remain inconsistent	Shared O-ISAC taxonomy, PRISMA-style ledgers, metric-governed templates
Hardware scalability	OPA/ORIS/PIC/photonic-THz demonstrations are promising but unevenly integrated	Co-design aperture, source, receiver, calibration, and packaging metrics
Channel modeling/evaluation	Fiber, FSO, VLC, and photonic-THz assumptions are often mixed	Scenario-tagged benchmark channels and plane-aware signal-quality reporting
Security/reliability	ML, sensing data, and narrow-beam control add new attack and failure surfaces	Threat-modeled evaluation with robustness and privacy fields
Deployment convergence	Application claims outpace cross-domain validation	Domain-aware pilots tied to reproducible evaluation workflows

TABLE XII
FIVE-ITEM PRIORITIZED RESEARCH AGENDA.

Pri.	Agenda item	Concrete deliverable	Risk if deferred
1	Metric-governed reporting	Required fields for rate, $\Delta r_{\min}$, σ_r /RMSE, CRB/FIM, Δz , OSNR/SNR plane	Persistent unsafe comparisons
2	Reproducible corpus workflow	Public ledgers, search strings, extraction sheets, and code for figures/tables	Weak systematic-review traceability
3	Cross-modality benchmarks	Fiber/FSO/VLC/photonic-THz scenario templates with uncertainty reporting	Non-transferable performance claims
4	Scalable hardware evidence	OPA/ORIS/PIC/photonic-THz prototypes evaluated with shared fields	Enabler claims remain isolated
5	Deployment validation	Domain pilots for infrastructure, indoor, vehicle, harsh OWC, and space/NTN cases	Application map remains aspirational

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